

CloudExify Cybersecurity Internship

Project 01

Network Penetration Testing Report

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➤ Introduction

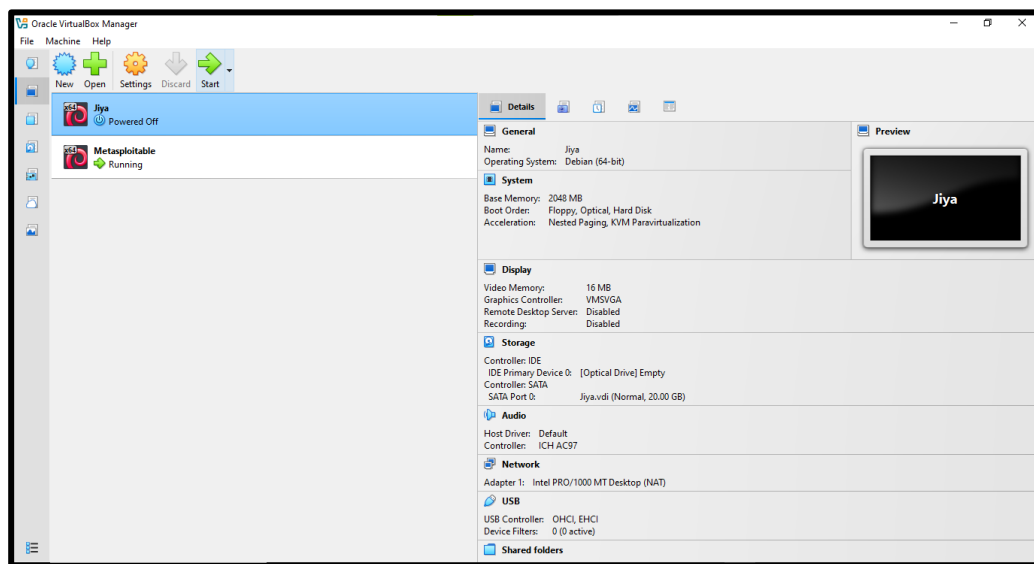
This project demonstrates a basic penetration testing workflow using VirtualBox, Kali Linux, Metasploitable and Nmap in a safe lab environment.

➤ Tools Used

- VirtualBox
- Kali Linux
- Metasploitable
- Nmap
- Wireshark

➤ VirtualBox Environment Setup

○ Screen shot



○ Purpose:

To set up the virtual lab environment using VirtualBox for cybersecurity testing.

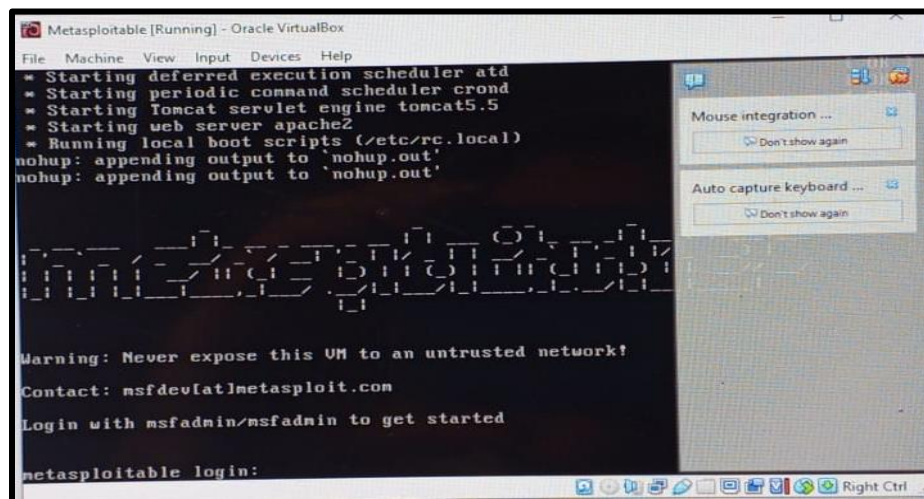
○ Description:

Oracle VirtualBox was configured with two virtual machines: Jiya (Kali Linux) and Metasploitable 2. These virtual machines were used to create a secure cybersecurity lab environment. VirtualBox Manager showing the configured virtual machines (Jiya – Kali Linux and Metasploitable 2).

➤ **Metasploitable Virtual Machine**

➤ **Metasploitable Login**

○ **Screen shot**



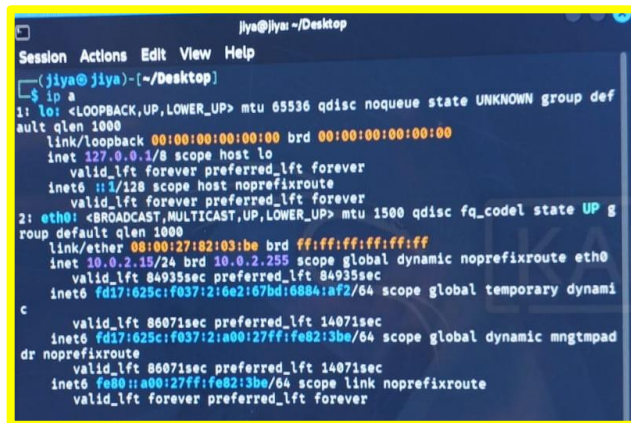
➤ **IP Address Discovery**

Before performing network communication and security scanning, it is necessary to identify the IP address of the target machine. The IP address acts as a unique identifier on the network and is required for commands such as ping and Nmap. In this project, the ip a command was used to discover the IP address of the Metasploitable virtual machine.

➤ **IP Address Identification (ip a)**

○ **Command Used:**

○ **Screen shot**



```
jiya@jiya: ~/Desktop
Session Actions Edit View Help
(jiya@jiya)-[~/Desktop]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:82:03:be brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 84935sec preferred_lft 84935sec
    inet6 fd17:625c:f037:216e2:67bd:6884:af2/64 scope global temporary dynami
c
        valid_lft 86071sec preferred_lft 14071sec
    inet6 fd17:625c:f037:21a00:127ff:fe82:3be/64 scope global dynamic mngtmpad
r noprefixroute
        valid_lft 86071sec preferred_lft 14071sec
    inet6 fe80::a00:127ff:fe82:3be/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

- **Purpose:**

To identify the IP address and network interface of the Metasploitable virtual machine for communication and network scanning.

- **Description:**

The ip a command was executed in the Metasploitable terminal to display the system's network configuration. The output shows the available network interfaces and the assigned IP address, which is used as the target for subsequent penetration testing activities.

- **Result:**

The IP address of the Metasploitable virtual machine was successfully identified using the ip a command. This IP address was then used for the Ping Test and Nmap scanning.

➤ **Ping Test**

- **Command Used:**

- **Screen shot**

```
jiya@jiya: ~/Desktop
Session Actions Edit View Help
(jiya@jiya)~-[~/Desktop]
$ ping 192.168.100.37
PING 192.168.100.37 (192.168.100.37) 56(84) bytes of data.
64 bytes from 192.168.100.37: icmp_seq=1 ttl=64 time=0.711 ms
64 bytes from 192.168.100.37: icmp_seq=2 ttl=64 time=0.786 ms
64 bytes from 192.168.100.37: icmp_seq=3 ttl=64 time=1.05 ms
64 bytes from 192.168.100.37: icmp_seq=4 ttl=64 time=0.941 ms
64 bytes from 192.168.100.37: icmp_seq=5 ttl=64 time=1.18 ms
64 bytes from 192.168.100.37: icmp_seq=6 ttl=64 time=1.22 ms
64 bytes from 192.168.100.37: icmp_seq=7 ttl=64 time=1.22 ms
64 bytes from 192.168.100.37: icmp_seq=8 ttl=64 time=0.703 ms
64 bytes from 192.168.100.37: icmp_seq=9 ttl=64 time=1.08 ms
64 bytes from 192.168.100.37: icmp_seq=10 ttl=64 time=1.24 ms
64 bytes from 192.168.100.37: icmp_seq=11 ttl=64 time=1.26 ms
64 bytes from 192.168.100.37: icmp_seq=12 ttl=64 time=1.25 ms
64 bytes from 192.168.100.37: icmp_seq=13 ttl=64 time=1.08 ms
64 bytes from 192.168.100.37: icmp_seq=14 ttl=64 time=1.31 ms
64 bytes from 192.168.100.37: icmp_seq=15 ttl=64 time=1.09 ms
64 bytes from 192.168.100.37: icmp_seq=16 ttl=64 time=1.06 ms
64 bytes from 192.168.100.37: icmp_seq=17 ttl=64 time=1.40 ms
64 bytes from 192.168.100.37: icmp_seq=18 ttl=64 time=1.08 ms
64 bytes from 192.168.100.37: icmp_seq=19 ttl=64 time=0.972 ms
64 bytes from 192.168.100.37: icmp_seq=20 ttl=64 time=0.584 ms
```

- Output
- Screen shot

```
jiya@jiya: ~/Desktop
Session Actions Edit View Help
64 bytes from 192.168.100.37: icmp_seq=24 ttl=64 time=0.911 ms
64 bytes from 192.168.100.37: icmp_seq=25 ttl=64 time=0.946 ms
64 bytes from 192.168.100.37: icmp_seq=26 ttl=64 time=0.843 ms
64 bytes from 192.168.100.37: icmp_seq=27 ttl=64 time=0.966 ms
64 bytes from 192.168.100.37: icmp_seq=28 ttl=64 time=0.944 ms
64 bytes from 192.168.100.37: icmp_seq=29 ttl=64 time=0.492 ms
64 bytes from 192.168.100.37: icmp_seq=30 ttl=64 time=0.843 ms
64 bytes from 192.168.100.37: icmp_seq=31 ttl=64 time=0.887 ms
64 bytes from 192.168.100.37: icmp_seq=32 ttl=64 time=0.559 ms
64 bytes from 192.168.100.37: icmp_seq=33 ttl=64 time=0.832 ms
64 bytes from 192.168.100.37: icmp_seq=34 ttl=64 time=0.925 ms
64 bytes from 192.168.100.37: icmp_seq=35 ttl=64 time=0.730 ms
64 bytes from 192.168.100.37: icmp_seq=36 ttl=64 time=0.492 ms
64 bytes from 192.168.100.37: icmp_seq=37 ttl=64 time=0.498 ms
64 bytes from 192.168.100.37: icmp_seq=38 ttl=64 time=0.424 ms
64 bytes from 192.168.100.37: icmp_seq=39 ttl=64 time=0.498 ms
64 bytes from 192.168.100.37: icmp_seq=40 ttl=64 time=0.736 ms
64 bytes from 192.168.100.37: icmp_seq=41 ttl=64 time=0.876 ms
64 bytes from 192.168.100.37: icmp_seq=42 ttl=64 time=1.26 ms
64 bytes from 192.168.100.37: icmp_seq=43 ttl=64 time=1.26 ms
64 bytes from 192.168.100.37: icmp_seq=44 ttl=64 time=1.31 ms
^C
— 192.168.100.37 ping statistics —
44 packets transmitted, 44 received, 0% packet loss, time 43270ms
rtt min/avg/max/mdev = 0.365/0.921/1.398/0.269 ms
```

- Purpose:

To check the network connectivity of the target machine and confirm that the host is online and reachable.

- Description:

Ping test is the first step of network reconnaissance. The `ping` command sends ICMP echo requests to the target IP. If a reply is received, it means the target machine is up and can communicate with the attacker. This step is performed to verify the lab setup before starting any exploit.

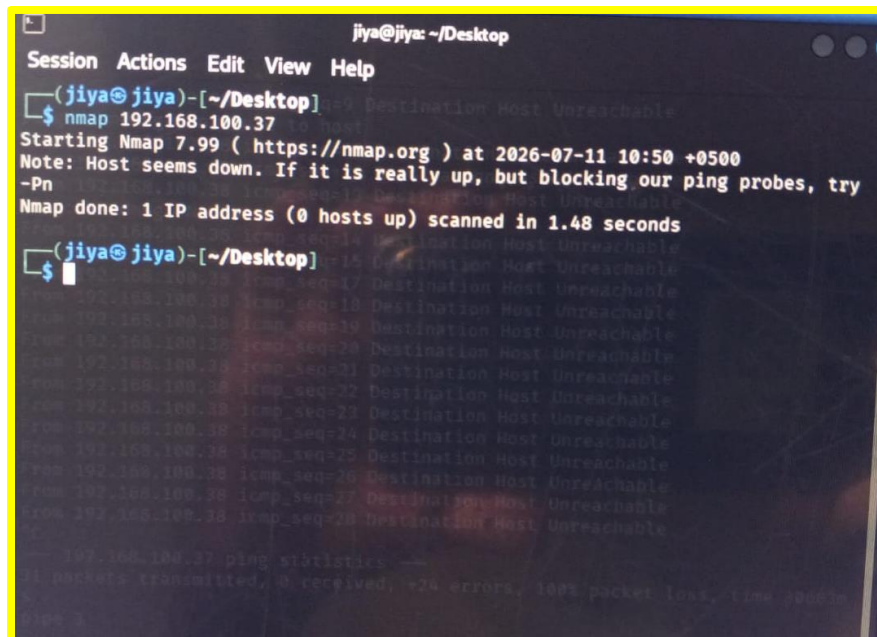
- Result:

Successful replies were received from target IP `192.168.100.37` with 0% packet loss. This confirmed that the target machine is online and the network is properly configured.

➤ Nmap Scan

- Command Used:

- Screen shot



```
jiya@jiya: ~/Desktop
Session Actions Edit View Help
(jiya@jiya)-[~/Desktop]
$ nmap 192.168.100.37
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-11 10:50 +0500
Note: Host seems down. If it is really up, but blocking our ping probes, try
-Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.48 seconds
(jiya@jiya)-[~/Desktop]
$
```

- Output

- Screen shot

```
jiya@jiya: ~/Desktop
Session Actions Edit View Help
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:C3:5F:A1 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.74 seconds
```

- Purpose:

To identify open ports, services, and their versions on the target machine in order to find a vulnerable service that can be exploited.

- Description:

Nmap is a network scanning tool. We used the `-sV` flag to detect service versions as well. This scan revealed which services are running on the target and where an attack could be performed.

- Result:

21/tcp - ftp vsftpd 2.3.4

22/tcp - ssh OpenSSH 4.7p1

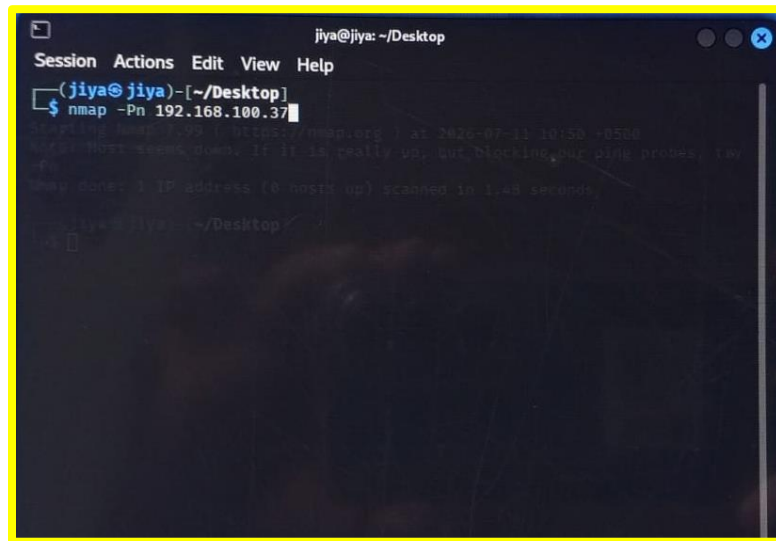
80/tcp - http Apache httpd 2.2.8

139/tcp - netbios-ssn Samba smbd 3.0.20

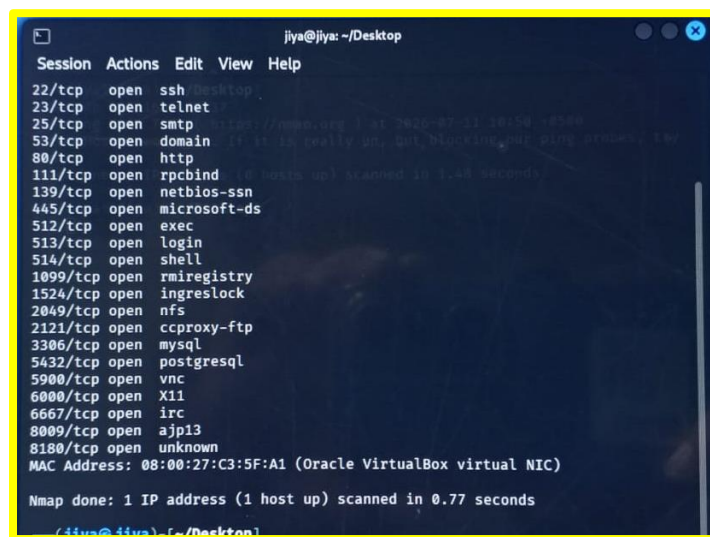
445/tcp - microsoft-ds Samba smbd 3.0.20

From these, `vsftpd 2.3.4` and `Samba` were found to be vulnerable and were exploited.

- Screen shot



- Screen shot



- Purpose:

To identify open ports and active services on the target machine.

- Description:

The `nmap -Pn 192.168.100.37` command was executed to scan the target host without performing host discovery. This allowed Nmap to identify open TCP ports and running services.

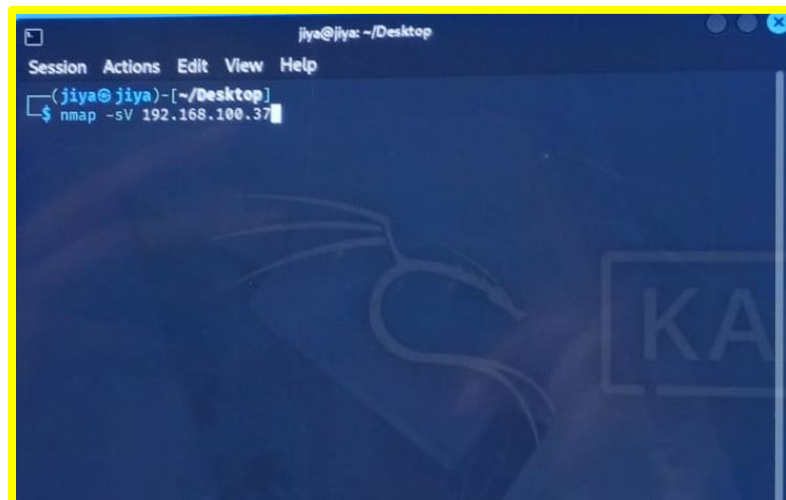
- Result:

The scan successfully detected multiple open ports including SSH (22), Telnet (23), HTTP (80), MySQL (3306), PostgreSQL (5432), and other services. The target host was confirmed to be active and accessible.

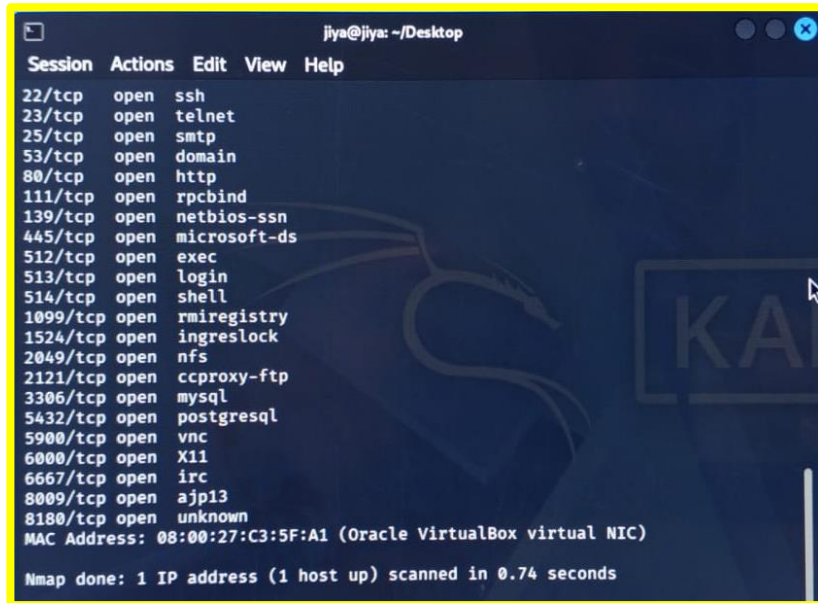
➤ Service Enumeration (nmap -sV)

- Command Used:

- Screen shot

A screenshot of a terminal window with a dark blue background and a yellow border. The window title is 'jiya@jiya: ~/Desktop'. The menu bar shows 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(jiya@jiya) - [~/Desktop]'. The command '\$ nmap -sV 192.168.100.37' has been entered and is followed by a cursor. In the background, there is a faint watermark of a hand holding a pen and the letters 'KA'.

- Output
- Screenshot



```

jiya@jiya: ~/Desktop
Session Actions Edit View Help
22/tcp open  ssh
23/tcp open  telnet
25/tcp open  smtp
53/tcp open  domain
80/tcp open  http
111/tcp open  rpcbind
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
512/tcp open  exec
513/tcp open  login
514/tcp open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:C3:5F:A1 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.74 seconds
  
```

- Purpose:

To identify the services and their versions running on the open ports of the target machine.

- Description:

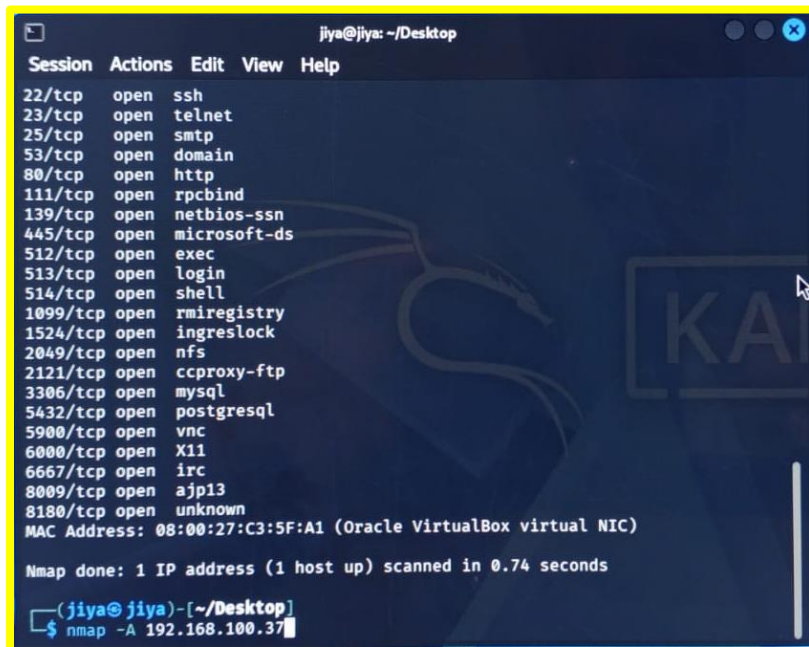
The nmap -sV command was executed to detect the versions of the services running on the target system. This information helps in understanding the software installed and identifying potential security weaknesses.

- Result:

The scan successfully identified multiple running services and their versions on the target machine, providing detailed information for further security assessment.

➤ Aggressive Scan (nmap -A)

- Command Used:
- Screen shot

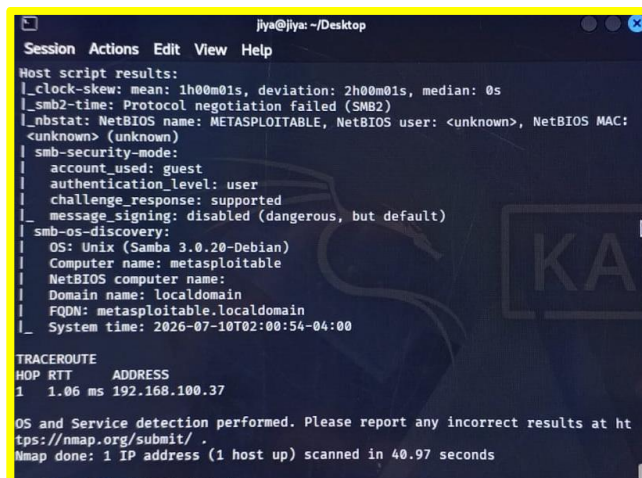


```
jiya@jiya: ~/Desktop
Session Actions Edit View Help
22/tcp open  ssh
23/tcp open  telnet
25/tcp open  smtp
53/tcp open  domain
80/tcp open  http
111/tcp open  rpcbind
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
512/tcp open  exec
513/tcp open  login
514/tcp open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:C3:5F:A1 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.74 seconds

(jiya@jiya)-[~/Desktop]
$ nmap -A 192.168.100.37
```

- Output
- Screenshot



```
jiya@jiya: ~/Desktop
Session Actions Edit View Help
Host script results:
|_clock-skew: mean: 1h00m01s, deviation: 2h00m01s, median: 0s
|_smb2-time: Protocol negotiation failed (SMB2)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC:
<unknown> (unknown)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_message_signing: disabled (dangerous, but default)
|_smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_System time: 2026-07-10T02:00:54-04:00

TRACEROUTE
HOP RTT ADDRESS
1 1.06 ms 192.168.100.37

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 40.97 seconds
```

○ Purpose

To perform a detailed scan of the target IP address (192.168.100.37) and gather comprehensive information about the target system.

○ Description

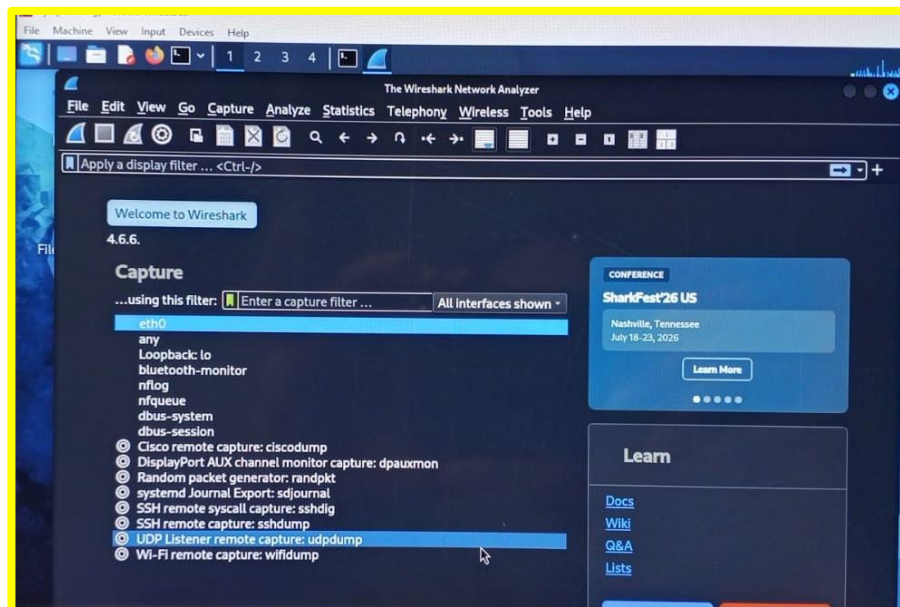
The `nmap -A 192.168.100.37` command was executed in Kali Linux to perform an aggressive scan. This scan enables operating system detection, service version detection, default script scanning, and traceroute. It provides detailed information about the target machine and its running services.

○ Results

The aggressive scan successfully identified the target host and displayed detailed information, including open ports, running services, service versions, operating system details (if detected), and additional information obtained through Nmap scripts. This helped in understanding the target system's network configuration and available services.

➤ Wireshark Environment

○ Screenshot



➤ Wireshark Packet Analysis

○ Purpose

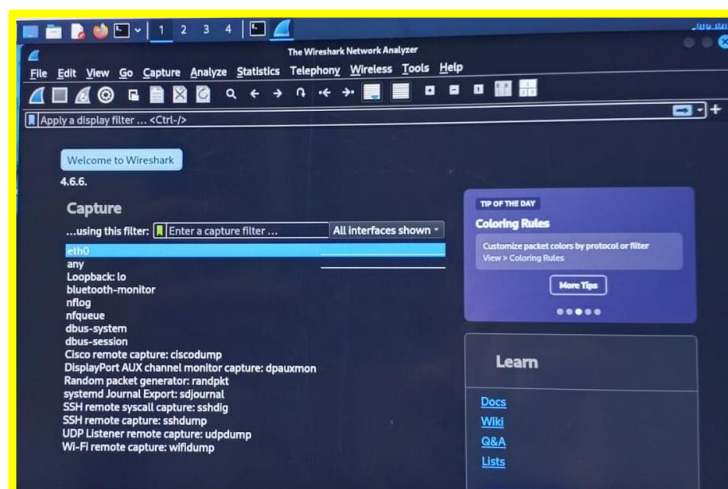
To capture and analyze live network traffic between Kali Linux and the Metasploitable virtual machine during the penetration testing process.

○ Introduction

Wireshark is a powerful network protocol analyzer used to monitor and inspect data packets transmitted over a network. It helps security professionals understand network communication, troubleshoot connectivity issues, and analyze traffic generated during penetration testing. In this project, Wireshark was used to capture packets exchanged between the Kali Linux and Metasploitable virtual machines.

○ Selecting eth0 Interface to Start Packet Capture

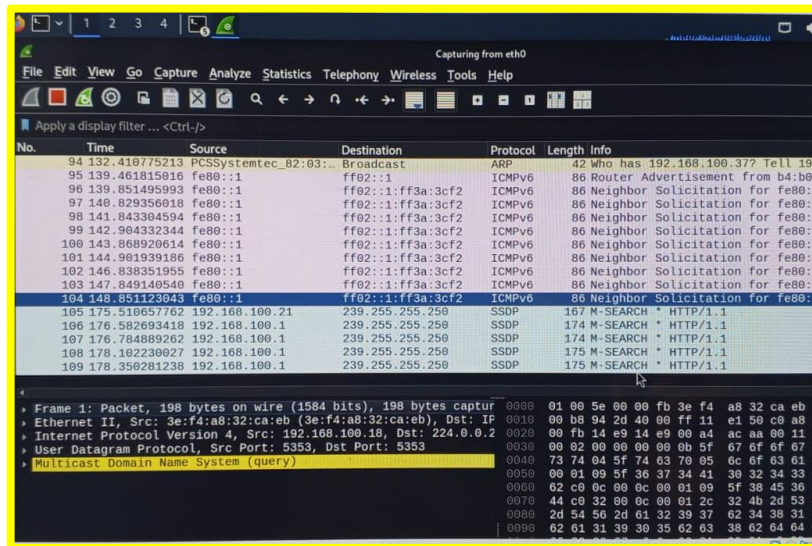
○ Screenshot



○ Description:

The Wireshark application was launched in Kali Linux, and the eth0 network interface was selected to capture live network traffic.

- Output
- Screenshot



- Description
Live network packets were successfully captured on the selected interface. The captured traffic confirmed successful communication between the Kali Linux and Metasploitable virtual machines.

○ Result

Wireshark successfully captured and displayed network packets generated during the penetration testing process. The captured traffic was analyzed to observe communication between the virtual machines and verify network connectivity.

➤ Vulnerability Identification

○ Purpose

To identify potential security weaknesses in the target system based on the open ports and running network services discovered during the Nmap scans.

○ Description

The results of the Nmap scans were analyzed to identify potential vulnerabilities associated with the detected services. Services such as FTP, Telnet, and HTTP use unencrypted communication, while database and file-sharing services should be properly secured to prevent unauthorized access. These observations help in understanding possible security risks within the target system.

- **Result**

The analysis identified multiple services that may pose security risks if they are not properly configured or secured. These findings provide a basis for recommending appropriate security improvements.

➤ **Identified Vulnerabilities Based on Open Ports and Services**

- **Table**

Port	Service	Potential Vulnerability
21	FTP	Unencrypted file transfer; usernames and passwords can be intercepted.
23	Telnet	Sends data in plain text, making it insecure for remote access.
80	HTTP	Web traffic is not encrypted and can be intercepted.
139 / 445	Samba	Misconfiguration may allow unauthorized access to shared files.
3306	MySQL	Database service should be restricted to authorized users only.
5432	PostgreSQL	Database service should not be publicly accessible.

➤ **Recommendations**

- **Purpose**

To provide security recommendations for reducing the risks identified during the penetration testing process.

- Description

Based on the identified open ports, running services, and potential vulnerabilities, the following security recommendations are suggested to improve the overall security of the target system.

- Result

Implementing these recommendations will help reduce security risks, improve system protection, and strengthen the network against common attacks.

➤ Recommended Security Measures:

1. Disable unnecessary open ports and services.
2. Replace Telnet with SSH for secure remote access.
3. Use HTTPS instead of HTTP to encrypt web traffic.
4. Keep all operating systems and services updated with the latest security patches.
5. Restrict access to database services such as MySQL and PostgreSQL using firewall rules and strong authentication.
6. Regularly monitor network traffic using Wireshark or other monitoring tools to detect suspicious activities.

➤ Conclusion

- Purpose

To summarize the activities performed and the outcomes achieved during the penetration testing project.

- Description

This project provided practical experience in performing basic penetration testing within a virtual lab environment. Kali Linux, Nmap, and Wireshark were used to identify active hosts, discover open ports, enumerate running services, and analyze network traffic. The collected information was then used to identify potential security vulnerabilities and recommend appropriate security improvements.

- Result

The penetration testing lab was completed successfully. The target machine was scanned, network traffic was captured, open ports and services were identified, and potential vulnerabilities were

documented. This project improved practical skills in network scanning, service enumeration, packet analysis, and security assessment.

